Individual member contributions

Project: Learning When to Search for Multi Agent Path Finding
Institution: IIT Roorkee
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Allocated role: Search gate, experimental analysis and visual demonstration

The project idea and the central question of when a learned MAPF policy should invoke search are shared contributions. My individual allocation covers the gate, evaluation and demonstration, together with the corresponding result interpretation and submission review.

1. Gate design and training methodology

Responsibility for explaining the sixteen confidence, local-risk and temporal features, logistic gate score and search threshold. The allocated review covers counterfactual eight-step rollout labels, validation-based threshold selection, search-hold duration and the difference between predicting short-term benefit and predicting episode failure.

2. Experimental protocol and baseline comparisons

Responsibility for the evaluation setup: agent counts, test seeds, disjoint instance generation, baseline methods and random-trigger controls. This includes reviewing the 140 test instances, the saved method/parameter runs, retrospective search-frequency matching and the limitations of exact equal-compute claims.

3. Results and interpretation

Responsibility for the success, external search-call, runtime, flowtime and makespan comparisons. Review coverage includes the reported call savings, the common solved subset, the gate-alone negative result, the role of the LaCAM fallback and why plain LaCAM remains faster on the tested maps. This scope includes checking tables and plots against the retained result files.

4. Visual simulation and benchmarking

Responsibility for the demonstration walkthrough, including seeded maps, controller selection, playback, gate scores, search-call counters and recorded compute time. The allocated review covers the three-run comparison table, CSV export, and explaining why playback speed and repeated timings on one map are different from algorithm runtime and multi-map success rates.

5. Results documentation and submission review

Responsibility for the experiment, results, limitations and future-work sections of the report and the second presentation segment. This includes reviewing `gate.py`, `train.py`, `experiments.py`, the demo interface and the consistency of the packaged results with Abhivansh's method explanation.

Primary deliverable scope: Gate methodology, evaluation evidence, result presentation and visual demonstration. These responsibilities are the agreed allocation for individual preparation and review.